text suggestions can affect a human writer's emotional tone. Arnold et al. [6] showed predictive text encourages predictable writing. Anderson et al. [4] and Laban et al. [61] found LLMs like ChatGPT helped users generate more detailed ideas, but outputs were less semantically distinct across users [87], and participants felt less responsible for their produced ideas. Recent work from Pan et al. [88] demonstrates language models can enhance outputs via feedback. However, methods like *Iterative Self-refinement scenarios*, using another language model as an evaluator, may result in reward hacking, where the model exploits the evaluator's flaws. For alignment training, it's crucial to consider the absolute desirability of each potential response, not just how responses compare to one another in terms of preference. Towards this in our work, we create pairs consisting of an initial LLM-generated response and its refined counterparts that by nature are more contrastive (or closely comparable). Our results show that such a pairing results in improved alignment and agreement during preference ranking.

3 DESIGN CONSIDERATIONS TO IMPROVE AI WRITING

"The secret to good writing is good editing.It's what separates hastily written, randomly punctuated, incoherent rants from learned polemics and op-eds, and cringe-worthy fan fiction from a critically acclaimed novel " [46]. In this section, we outline the design principles and desiderata that guided our approach to improving AI writing through textual edits.

Design Principle 1: Develop a comprehensive edit taxonomy grounded in expert writing practices. This principle emphasizes creating a comprehensive taxonomy of edit categories [33] rooted in an expert editor and writer's practices. Prior work has shown that experts and novices define revising in very different ways with experts attending more systematically to different aspects of the text than novices [9, 100, 106, 113]. By developing such a taxonomy, we aim to provide an approach to analyzing and enhancing LLM-generated text. It also allows for a more granular understanding of the specific areas where AI writing may fall short and enables targeted improvements. Sommers [106] found that "experienced writers have a second objective; a concern for their readership". Grounding the taxonomy in expert writing practices ensures that the edits align with the standards of high-quality writing and are acceptable to its readers. Finally, this principle also acknowledges the complexity of the editing process, recognizing that different categories of edits may be required at various levels of the text, from sentence-level corrections to broader structural changes [47, 106, 115].

Design Principle 2: Leverage edits to balance both meaning preservation and substantive semantic changes. Preserving the core meaning and intent of the original text is crucial to maintaining coherence and faithfulness to the initial ideas. On the other hand, introducing substantive semantic changes is often required to adhere to the quality and characteristics of good writing. Prior work on edit taxonomies focuses on low-level syntactic operations [33] or semantic edits [22, 62, 115] tailored to specific websites like Wikipedia. LLM-generated text often benefits from syntactic edits. These edits (primarily meaning preserving) enhance readability by diversifying sentence structures, expanding vocabulary choices, and minimizing repetitive phrasing. Consequently, semantic edits (both meaning preserving and changing) in AI writing are important for enhancing specificity or reducing unnecessary flourishes and clichés that can otherwise obscure meaning. Our methodology aims to navigate the tension between maintaining original meaning and introducing necessary improvements to mitigate AI-specific writing quirks.

Design Principle 3: Utilize edits as a mechanism for enhancing human-AI alignment in writing. Current AI writing systems are developed using pre-trained language models (LMs) refined through human interaction, employing supervised learning and reinforcement learning (RL) techniques. Reinforcement learning from human feedback (RLHF) [121] is a key approach, transforming human input into training data to guide LMs toward desired outcomes. The most common type of feedback used with RLHF is binary preferences between pairs of examples sampled from one or more Language Models [19]. However, a learned preference ordering can fail to converge to the true one when the desirability of examples depends on noise [42]. Following recent work in preference learning [25, 30], we evaluate edits as a mechanism for enhancing alignment. An LLM-generated response that has been edited typically contains fewer undesirable traits and can be paired with the original LLM-generated response for preference ranking ($edited > original$).While contemporaneous works [25, 30] have conducted preliminary efforts to incorporate edits for improved preference data collection, we evaluate this approach in the context of creative writing.

4 LARGE SCALE DATA COLLECTION PROCESS

We aim to create a valid test-bed to evaluate the quality of LLM-generated text on realistic writing tasks that require creative skill. We follow a three-step approach illustrated in Figure 2: (1) First we select original paragraphs of human-written text from trusted publication venues, (2) Second we reverse-engineer each of these paragraphs into a writing instruction. Because each instruction originates from an existing human-written paragraph within a piece of creative writing, this simulates real-world writing situations. (3) Third, we prompt several LLMs to generate responses to each of the writing instructions. In the following subsections, we first detail each of these steps, and then describe the formative study we conducted to develop a taxonomy of idiosyncrasies in LLM-generated text.

4.1 Collecting Instruction and Response Pairs

To curate source material, we select five well-regarded [3] publication venues, listed in Table 1, that publish pieces in different domains ranging from fiction to food writing and internet advice. For each venue, we manually extract between 100-700 pieces of writing and isolate individual paragraphs. We then manually review these paragraphs, ensuring they are long enough and can stand alone as coherent pieces of writing without requiring additional context. In total, we selected approximately 1200 paragraphs following this procedure. The Literary Fiction genre has a larger representation (80%) in our selection, while the creative non-fiction genres have a smaller representation.

[3]https://www.pewresearch.org/politics/2012/09/27/section-4-demographics-and-political-views-of-news-audiences/

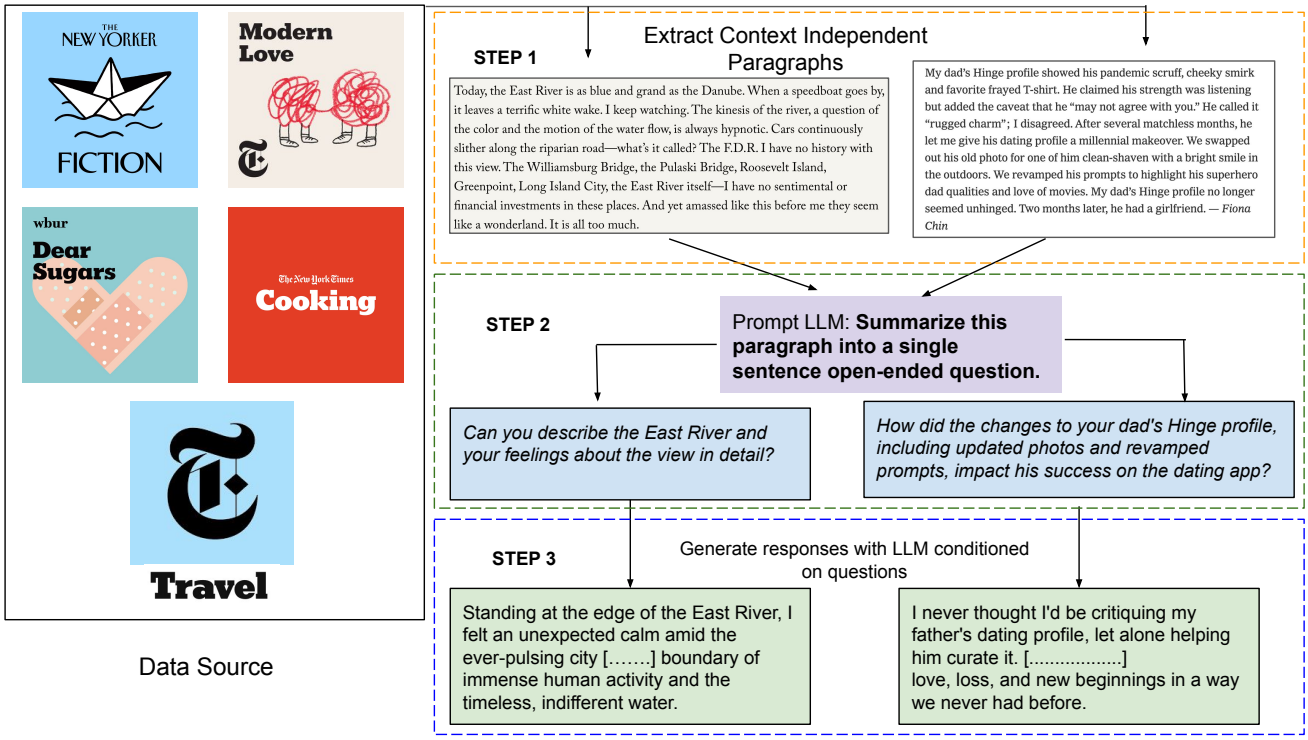

Figure 2: The pipeline for data creation. Step 1) Extracting context-independent paragraphs from our respective sources Step 2) Using an LLM to automatically generate instructions for corresponding human-written text Step 3) Use the generated instructions grounded in real-world writing to elicit responses from LLMs to create <instructions,response> pairs

Next, we follow Li et al. [67]'s approach of *Instruction Backtranslation* to automatically generate instructions corresponding to each of the selected paragraphs. Specifically, we prompt an LLM (i.e., GPT4o) to summarize each paragraph into an open-ended question. Questions obtained through back-translation (see examples in Table 1) can be interpreted as realistic writing instructions. We manually verified the generated instructions, filtering out questions that were ill-formed or overly specific, yielding a total of 1,057 writing instructions.

Finally, we use the generated instructions to elicit responses from three state-of-the-art LLMs: OpenAI's GPT-4o [83], Anthropic's Claude-3.5-Sonnet [5], and Meta's Llama 3.1-70b [30]. Each LLM is used to generate responses to one-third of our instruction data. We ensure that each LLM responds to instruction across all domains in equal proportion. To generate high-quality responses, we provide each LLM with the writing instruction, as well as the genre and source, and instruct it to adhere to the style of the venue. The prompt further specifies: *"Try your best to be original, avoiding clichés or overused tropes. Do not use ornamental language and focus on nuance, simplicity, and subtext"* (See Prompt in Appendix A Table 14). Through this process, we obtain 1,057 writing <instructions, response> pairs, with responses averaging 205 words. This collection of instructions and LLM-generated responses serves as the foundation for the three studies we conducted: the formative study, the full-scale editing annotation, and the preference annotation.

4.2 Formative study: formulating the taxonomy for fine-grained edits

Our formative study observed writers with copy-editing experience as they edited LLM-generated text in the creative writing domain. We aimed to identify common edit categories. The study consisted of three phases. First, participants were individually briefed via video conference on the study's objectives. Next, they accessed a web application (Figure 10) to view <instruction, response> pairs from our dataset (Section 5). For each sample, participants highlighted problematic response spans, suggested rewrites, and tagged each span with a free-form category to characterize the issue. We recruited eight participants for the formative study, with each completing annotations for 25 samples.

For participant selection in our formative study, we limited involvement to individuals with established expertise in creative writing. Participants were required to have completed a Master of Fine Arts (MFA) in Creative Writing and were recruited through mailing lists from MFA writing programs in the United States. This aligns with prior work from Chakrabarty et al. [15], using the Consensual Assessment Technique [2], which emphasizes the importance of recruiting domain experts. During the initial video call, we confirmed participants' familiarity with copy-editing and informed

Domain	Source & Genre	Example Seed Paragraph with Generated Instruction	#N
Fiction	The NewYorker (Literary Fiction)	The sunset is a red-gold rumpus on the western sky. It has rained. The crow tosses itself from branch to branch, pole to pole, glistening on its pace, and she follows. They are soon far from where they began, streets unfamiliar to her, an older part of town,[..... ....]. A man reading. Old Christmas tree in a corner. It feels secret. The sky is clearing overhead. She feels secret, too. She feels tremendous. Instruction : **Can you describe a vivid scene at sunset that transitions into nighttime, incorporating elements of nature, urban surroundings, and personal observations?**	815
Creative Non-Fiction	NYTimes (Travel Writing)	Prague, the Czech capital, is finding a new balance between preserving its past and embracing the future, improving many of its important historic sites while making striking additions to its skyline. [.....] Stop by for a coffee, hit up one of the many great new bakeries or visit a charismatic old beer hall as you explore a city that is clearly entering its prime. Instruction : **How is Prague balancing its historical preservation with modern development while enhancing local amenities and vibrant neighborhoods outside Old Town?**	110
	NYTimes (Food Writing)	The origins of the fruit sandwich are believed to go back to Japan's luxury fruit stores & the fruit parlors attached to them. This version comes from Yudai Kanayama, a native of Hokkaido who runs the restaurants the Izakaya NYC and Dr Clark in New York. [.....] .The sandwich looks like dessert but isn't, or not exactly; it makes for a lovely little meal that feels slightly illicit, as if for a moment there are no rules Instruction : **How did Yudai Kanayama reinvent the traditional Japanese fruit sandwich to create a unique culinary experience?**	83
	NYTimes (Personal Essay)	My dad's Hinge profile showed his pandemic scruff, cheeky smirk and favorite frayed T-shirt. He claimed his strength was listening [.....] We revamped his prompts to highlight his superhero dad qualities and love of movies. My dad's Hinge profile no longer seemed unhinged. Two months later, he had a girlfriend. Instruction : **How did the changes to your dad's Hinge profile, including updated photos and revamped prompts, impact his success on the dating app?**	19
	Dear Sugar (Internet Advice)	What is a prestigious college? What did attending such a school allow you to believe about yourself? What assumptions do you have about the colleges that you would not describe as prestigious? What sorts of people go to prestigious colleges and not [.....] I believe our early experiences and beliefs about our place in the world inform who we think we are and what we deserve and by what means it should be given to us. Instruction : **How do your beliefs and assumptions about educational privilege and the type of schools people attend shape your current view of yourself and others?**	30

Table 1: Venues of source material used to extract real-world examples of creative writing, along with example seed paragraphs and the generated instructions and the number of samples per genre.

them they would edit LLM-generated texts. We recruited participants through UserInterviews [4], a professional freelancing platform, paying $75 USD for study completion. Editing a response took 4-6 minutes, with all participants finishing within two hours. Table 2 shows diverse demographics and professional backgrounds of the recruited creative writers.[5] In total, the eight participants edited 200 paragraphs, annotating roughly 1,600 edits attributed to 50 distinct initial categories. We used this data as the foundation for our next analysis, which aimed to develop a taxonomy for categorizing edits.

[4]https://www.userinterviews.com

[5]The research was conducted at an institution without a formal IRB approval process. However, an Ethical Practices team reviewed the work and study protocols. No personally identifiable information (PII) was collected or shared during data collection, and participants were offered compensation regardless of study completion.